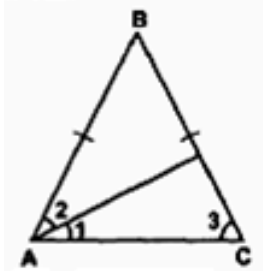
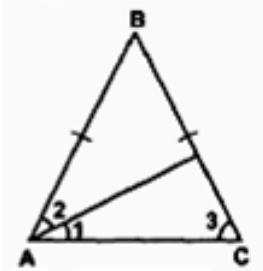
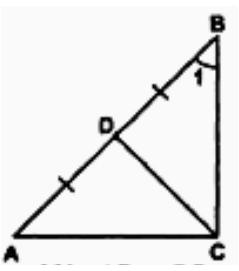
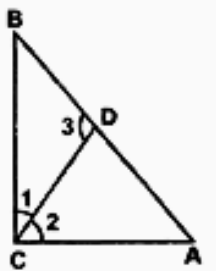
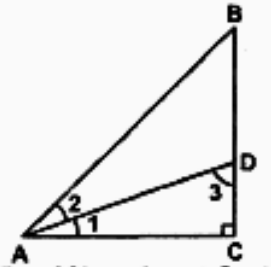
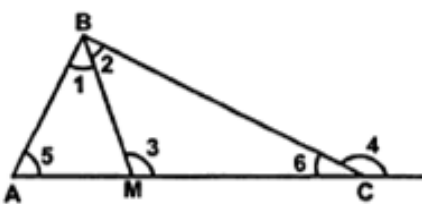
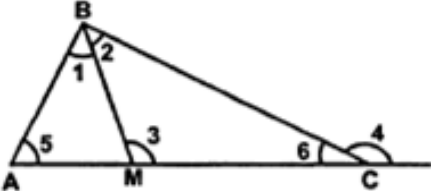
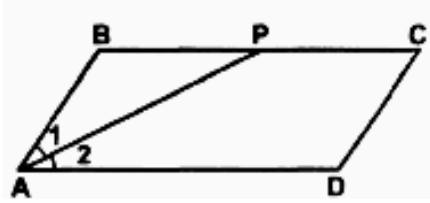
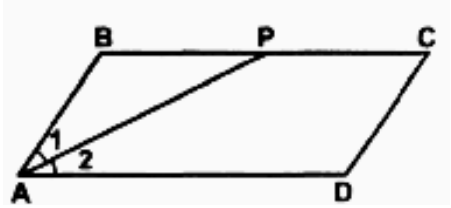
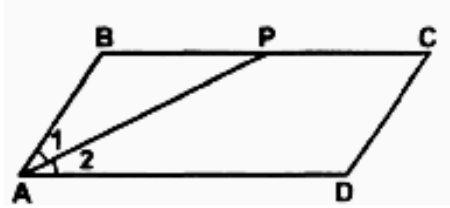
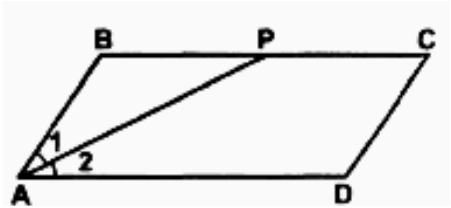
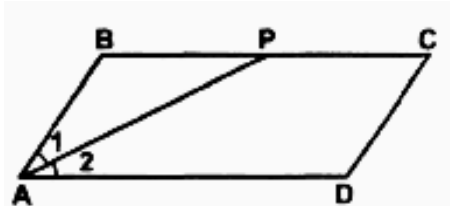
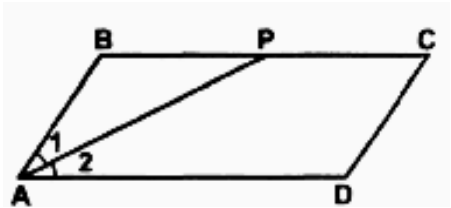
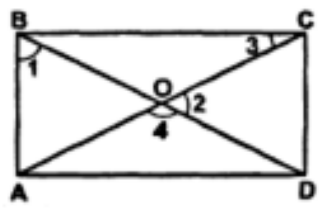
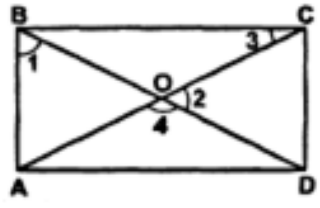
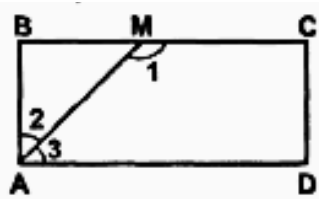
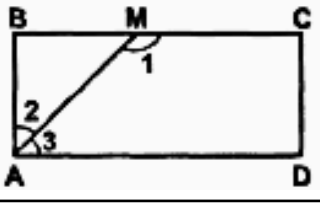
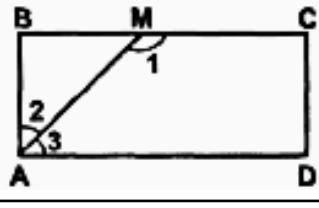


ВИДЫ ТРЕУГОЛЬНИКОВ. БИССЕКТРИСА, МЕДИАНА И ВЫСОТА ТРЕУГОЛЬНИКА

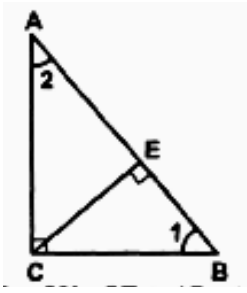
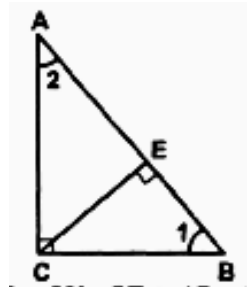
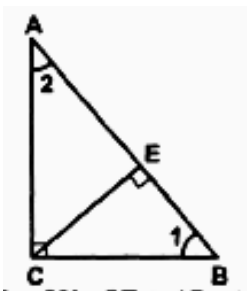
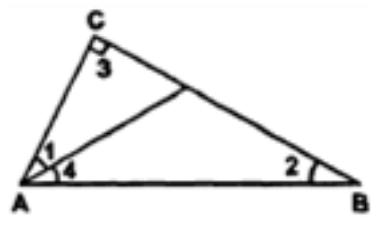
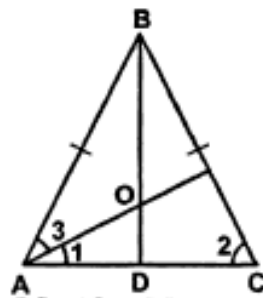
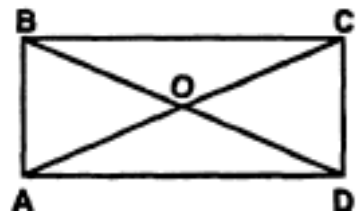
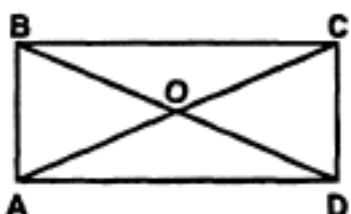
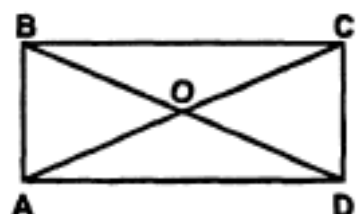
ҮШБҰРЫШТЫҢ ТҮРЛЕРІ. ҮШБҰРЫШТЫҢ БИССЕКТРИСАСЫ, МЕДИАНАСЫ ЖӘНЕ БИІКТІГІ

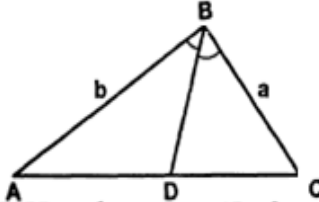
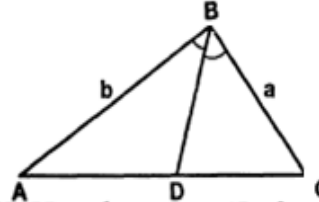
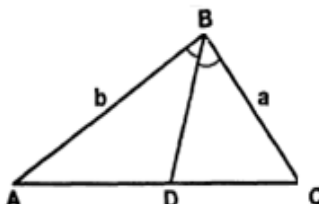
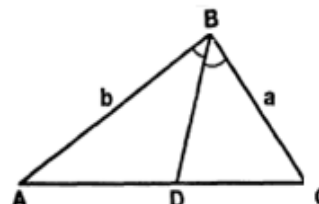
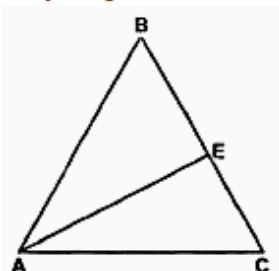
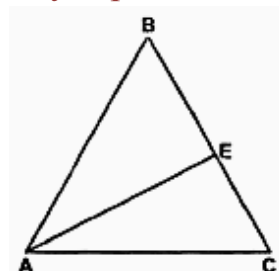
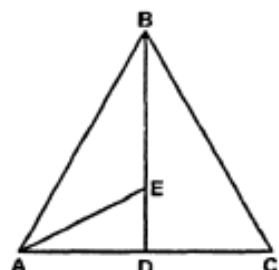
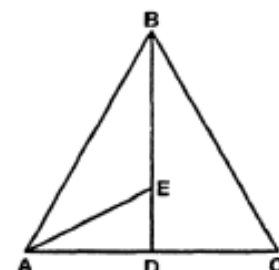
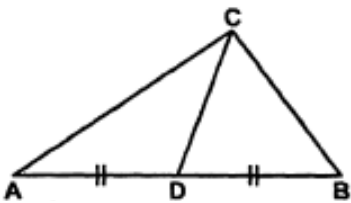
1 уровень/деңгей

1.	$AB = BC$, $\angle 1 = \angle 2$, $\angle 1 + \angle 3 = 45^\circ$ Найти/табу керек: $\angle B$	2.	$AB = BC$, $\angle 1 = \angle 2$, $\angle B = 80^\circ$ Найти/табу керек: $\angle 1 + \angle 3$
			
3.	$\angle ACB = 90^\circ$, $AD = DB$, $\angle 1 = 50\% \angle ACB$ Найти/табу керек: $\angle CDB$	4.	$\angle ACB = 90^\circ$, $\angle 1 = \angle 2$ $\angle 3 = 105^\circ$ Найти/табу керек: $AC : AB$
			
5.	$\angle ACB = 90^\circ$, $\angle 1 = \angle 2$ $AB = 2BC$ Найти/табу керек: $\angle 3$	6.	$\angle 1 = \angle 2$, $\angle 6 = 10^\circ$, $\angle 3 = 100^\circ$ Найти/табу керек: $\angle 5$
			
7.	$\angle 1 = \angle 2$, $\angle 4 = 135^\circ$, $\angle 3 = 100^\circ$ Найти/табу керек: $\angle ABC$	8.	$\angle 1 = \angle 2$, $BP = 7$, $PC = 3$ Найти/табу керек: P_{ABCD}
			

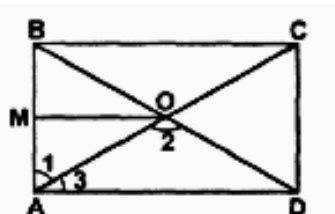
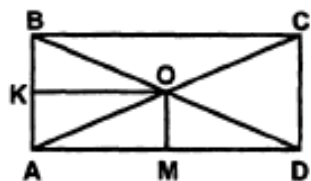
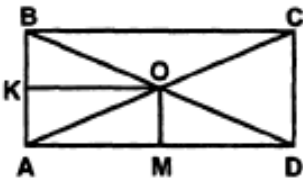
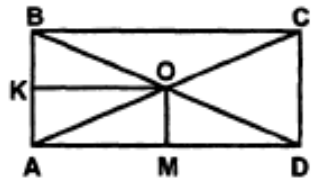
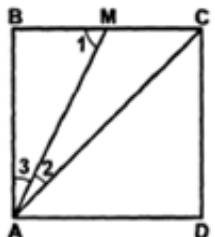
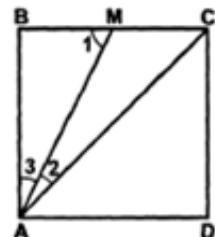
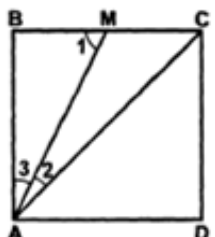
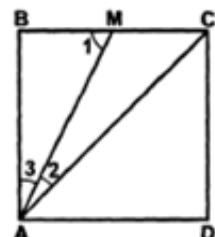
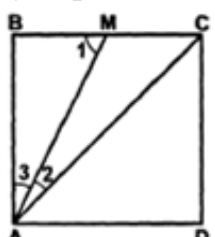
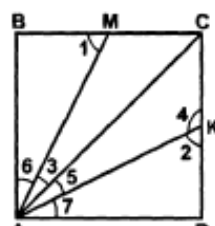
9.	$\angle 1 = \angle 2, AP = 15, DC = 9$ Найти/табу керек: P_{ABP}		10.	$\angle 1 = \angle 2, P_{ABCD} = 50,$ $PC = AD - 8$ Найти/табу керек: AD	
11.	$\angle 1 = \angle 2, P_{ABCD} = 20,$ $AD = 3PC$ Найти/табу керек: AD		12.	$\angle 1 = \angle 2, AD = AB + 2,$ $AB = PC - 1$ Найти/табу керек: P_{ABCD}	
13.	$\angle 1 = \angle 2, P_{ABCD} = 26,$ $AP = AB + 3, AD = AP$ Найти/табу керек: P_{APCD}		14.	$\angle 1 : \angle 3 = 7 : 2$ Найти/табу керек: $\angle 4$	
15.	$AC : CD = 2 : 1$ Доказать, что/дәлелдеу керек: $\angle 1 = \angle 2$		16.	$AB = BM$ Найти/табу керек: $\angle 1$	
17.	$AB = BM, \angle 1 + \angle D = 225^\circ,$ $AD = 10$ Найти/табу керек: $AB + MC$		18.	$\angle 3 = \angle 2, BM = 3, MC = 7$ Найти/табу керек: P_{ABCD}	

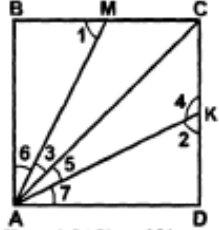
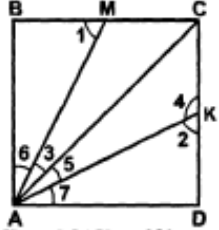
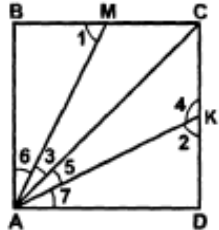
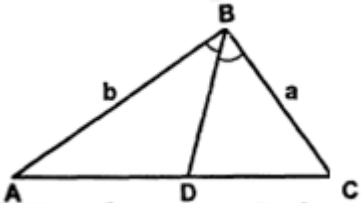
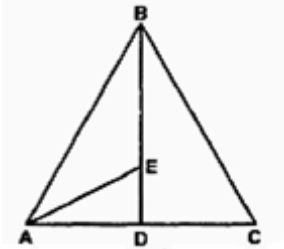
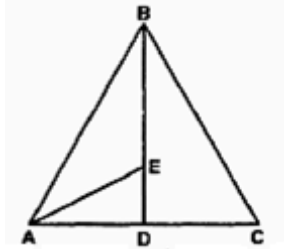
2 уровень/деңгей

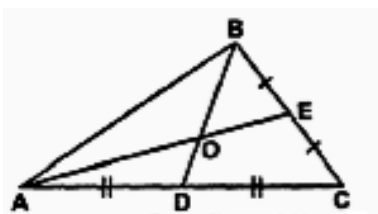
19.	$\angle ABC = 90^\circ, CE \perp AB, \angle 1 = 2\angle 2$, $AC + CE = 3$ Найти/табу керек: CE	20.	$\angle ABC = 90^\circ, CE \perp AB, \angle A = 30^\circ$, $AB + BC = 9$ Найти/табу керек: BE
			
21.	$\angle ABC = 90^\circ, CE \perp AB$, $\angle ECB = 30^\circ, BE = 1$ Найти/табу керек: AE	22.	$\angle 3 = 90^\circ, \angle 1 = \angle 4$, $\angle CAB = \angle 2 + 70^\circ$ Найти/табу керек: $\angle 1$
			
23.	$AB = BC, \angle 1 = \angle 3, AC \perp BD$ $\angle AOD - \angle 2 = 21^\circ$ Найти/табу керек: $\angle B$	20.	$P_{AOD} = 18, AC + DB = 22$ Найти/табу керек: BC
			
21.	$P_{ACD} = 49, P_{ABCD} = 62$ Найти/табу керек: AO	22.	$P_{COD} = 30, AC + BD = 40$ Найти/табу керек: $\angle AOD$
			
23.	BD – биссектриса, $AB = 2$, $BC = 8, AD = 1$ Найти/Табу керек: AC	24.	BD – биссектриса, $AB = 4$, $BC = 12, AC = 20$ Найти/Табу керек: AD, DC

			
25.	BD –бісектриса, $AB + BC = 12$, $DC = 8$, $AD = 2$ Найти/Табу керек: AB	26.	BD –бісектриса, $BC - AB = 3$, $DC = 8$, $AD = 6$ Найти/Табу керек: BC
			
27.	AE –бісектриса, $AB = BC$, $AB = 5$, $AC = 6$ Найти/Табу керек: BE , CE	28.	AE –бісектриса, $AB = BC$, $AB \cdot EC = 4$ Найти/Табу керек: $AC \cdot BE$
			
29.	$AB = BC$, $BD \perp AC$, AE – бісектриса, $BE : ED = 17 : 15$, $AC = 60$ Найти/Табу керек: P_{ABC}	30.	$AB = BC$, $BD \perp AC$, AE – бісектриса, $BE : ED = 13 : 12$, $P_{ABC} = 250$ Найти/Табу керек: AB
			
31.	$\angle ACB = 90^\circ$, $AD = DB$, $CD = 2$ Найти/Табу керек: AB		
			

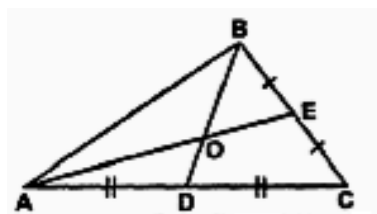
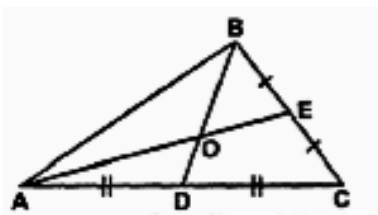
3 уровень/деңгей

32. $AM = MB, \angle 1 + \angle 2 = 198^\circ$ Найти/Табу керек: $\angle 3$ 	33. $OM \perp AD, OM = 7, AD = 17$ Найти/Табу керек: P_{ABCD} 
34. $OM \perp AD, P_{OMD} = 18, P_{ABCD} = 48$ Найти/Табу керек: AC 	35. $OM \perp AD, OK \perp AB, P_{ABCD} = 100$ Найти/Табу керек: $OM + OK$ 
36. $ABCD$ – квадрат, $\angle 1 = 72^\circ$ Найти/Табу керек: $\angle 2$ 	37. $ABCD$ – квадрат Доказать, что/Дәлелдеу керек: $\angle 1 - \angle 2 = 45^\circ$ 
38. $ABCD$ – квадрат, $AM = 2BM$ Найти/Табу керек: $\angle 2$ 	39. $ABCD$ – квадрат, $\angle 2 : \angle 1 = 1 : 4, AM = 3$ Найти/Табу керек: BM 
40. $ABCD$ – квадрат, $\angle 1 : \angle 2 = 11 : 2$ Найти/Табу керек: $\angle 3$ 	41. $ABCD$ – квадрат, $BM = DK, \angle MAK = 62^\circ$ Найти/Табу керек: $\angle 4$ 

42.	$ABCD$ – квадрат, $\angle 1 = 75^\circ$, $\angle 2 = 63^\circ$. M и K точки, принадлежащие сторонам BC и CD. Найти $\angle MAK$ M және K – BC және CD қабырғаларына тиісті нүктелер. Табу керек: $\angle MAK$ 	43. $ABCD$ – квадрат, M и K- точки, принадлежащие сторонам BC и CD. $\angle 1 : \angle 5 = 2 : 1$, $\angle 3 = 21^\circ$ Найти: $\angle 2$ M және K BC және CD қабырғаларына тиісті нүктелер. $\angle 1 : \angle 5 = 2 : 1$, $\angle 3 = 21^\circ$ Табу керек: $\angle 2$ 
44.	$ABCD$ – квадрат, $\angle 3 = 4\angle 5$, $\angle 7 = 7\angle 6$ Найти/Табу керек: $\angle 2$ 	45. BD – биссектриса, $AD=7$, $DC=5$, $P_{ABD} - P_{BCD} = 6$ Найти/Табу керек: AB 
46.	$AB = BC$, $BD \perp AC$, AE – биссектриса, $BE : ED = 17 : 15$, $AC = 60$ Найти/Табу керек: P_{ABC} 	47. $AB = BC$, $BD \perp AC$, AE – биссектриса, $BE : ED = 13 : 12$, $P_{ABC} = 250$ Найти/Табу керек: AB 
48.	$CE = BE$, $AD = DC$, $AE = 9$, $BD = 12$ Найти/Табу керек: $AO + DO$	49. $CE = BE$, $AD = DC$, $OE + OD = 2$ Найти/Табу керек: $AE + BD$



50. $CE = BE, AD = DC, AE = 6,$
 $BD = 9, AC = 12$
 Найти/Табу керек: P_{AOD}



51. $CE = BE, AD = DC, OD = 1,$
 $\angle ABC = 90^\circ$
 Найти/Табу керек: $BO + AC$

